

SAMPLE RATIO MISMATCH

Sample ratio between control and treatment
is not as designed

SAMPLE RATIO MISMATCH

- 1 (design ratio) vs 1.01 (observe ratio)
- p-value < 0.05
- Highly unlikely to observe such a ratio (or more extreme) conditioned on the design ratio

SAMPLE RATIO MISMATCH

Things get complicated when assigning users
to different groups

- tests with ramping up plans

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- running multiple tests in parallel

SAMPLE RATIO MISMATCH

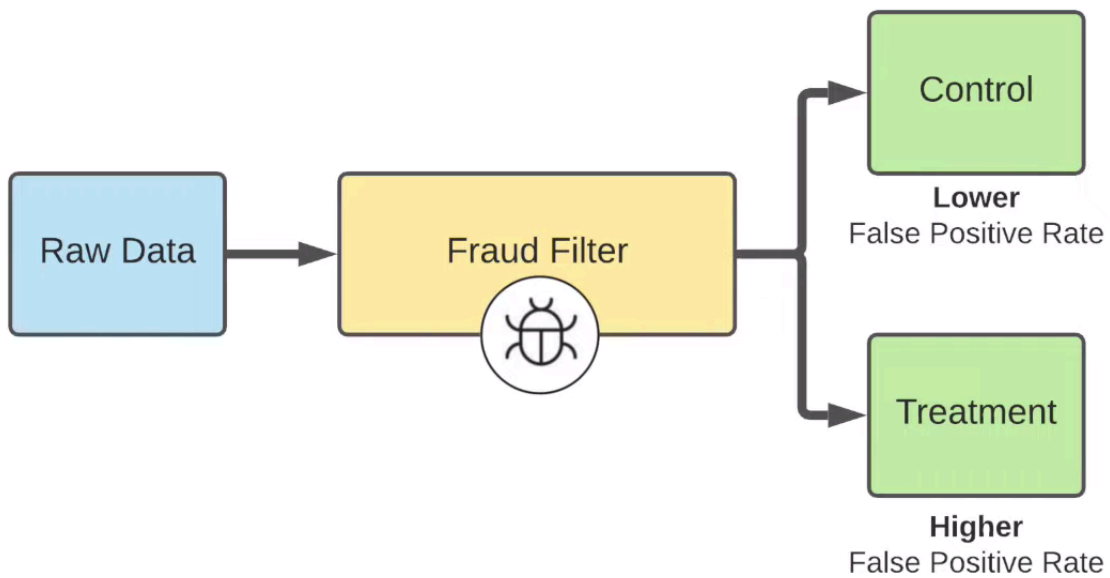
Things get complicated when assigning users to different groups

- tests with ramping up plans
- running multiple tests in parallel
- segmentation is based on attributes that can change

SAMPLE RATIO MISMATCH

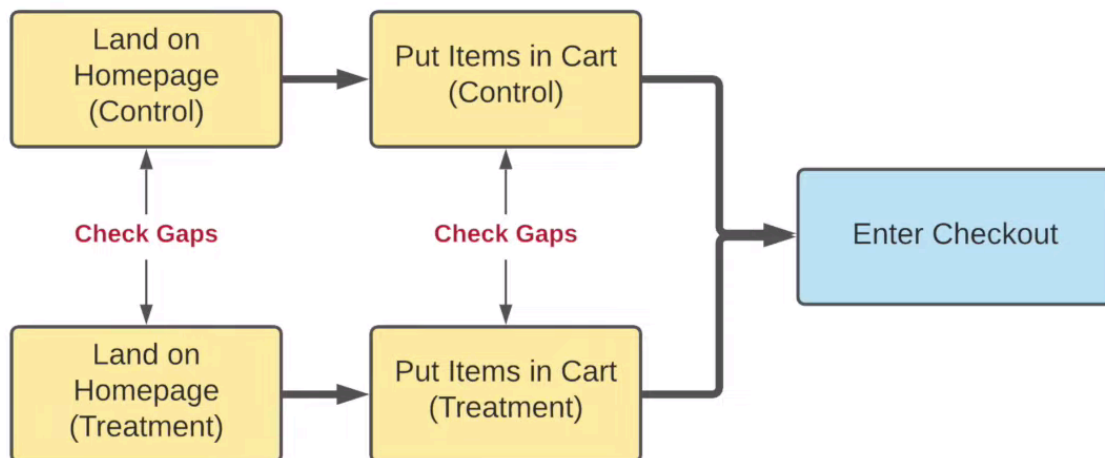
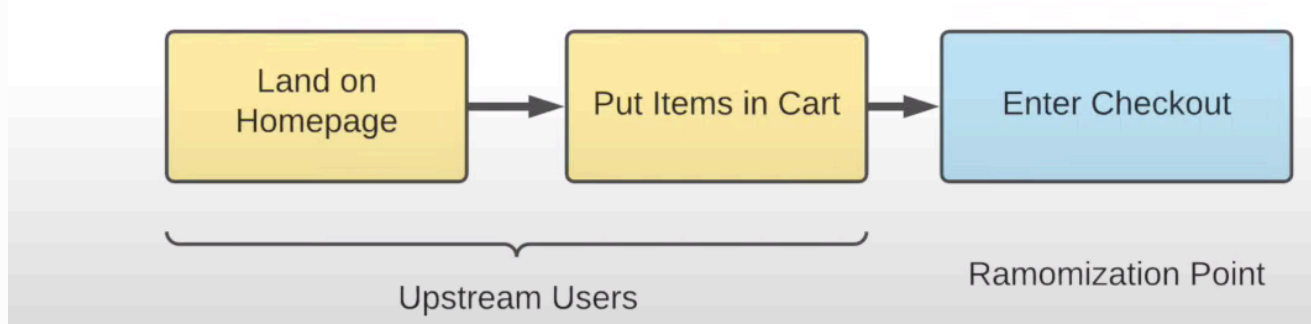
Causes

- bugs in assigning users to different groups
- the way results are processed



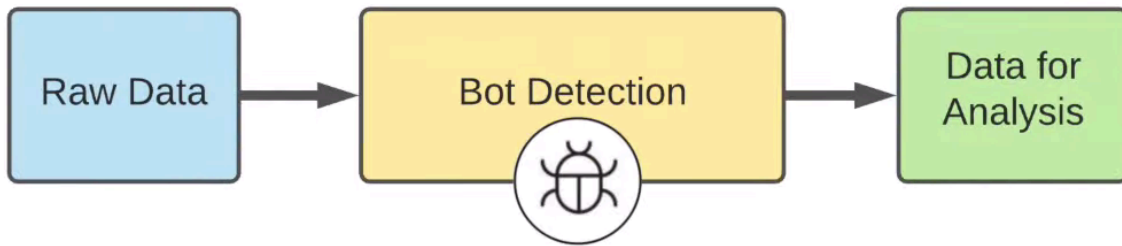
Debugging

- Gap upstream of the randomization point



Debugging

- Gap upstream of the randomization point
- Variant assignment



Debugging

- Gap upstream of the randomization point
- Variant assignment
- Data processing pipeline
- Segments of population

VIOLATION OF SUTVA

Stable unit treatment value assumption
(SUTVA)

- randomization units are **independent**
and **no interaction** between them

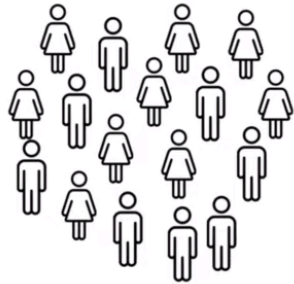
Social networks

- e.g. Facebook and LinkedIn
- User's behaviors are influenced by others in their social networks

Two-sided markets

- e.g. Ebay, Uber and Lyft
- Resources are shared between groups

VIOLATION OF SUTVA



**Treatment
Demand**



**Control
Demand**



**Treatment
Supply**



**Control
Supply**

Solution

- predict where the interference will happen
- isolate control and treatment units

Solution

- when prediction is difficult
- monitor and detect it

Novelty effect

- used a lot at the beginning
- then abandoned it

*Novelty effect:
Some users love new features and use it more*

*Primacy effect:
Other users hate it and use it less*

Solution

- monitor it and quantify it
- filter out novelty and primacy effects
when evaluating treatment effect

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- **Misunderstanding stats Concepts**

- **Assuming validity of results without sanity checks**

- Stop collecting data when the test comes out significant
- Result is **inaccurate**

When calculate duration of an experiment

- statistical power
- significance level
- day of week effect
- seasonality
- etc

Stop experience earlier than designed

- Estimated treatment effect $\neq$ true treatment effect

Goal of A/B tests

- result is **reproducible**

Stop tests prematurely

- result won't be reproducible